In [1]:
```
import pandas as pd
```

In [2]:
```
df = pd.read_csv(r"C:\Users\abhis\Downloads\house_prices_practice.csv")
```

In [3]:
```
df.describe()
```

Out[3]:
<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>Id</th> <th>OverallQual</th> <th>GrLivArea</th> <th>GarageCars</th> <th>TotalBsmtSF</th> <th>YearBuilt</th> <th>FullBath</th> <th>BedroomAbvGr</th> <th>LotArea</th> <th>SalePrice</th> </thead> <tbody> <th>count</th> <th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th> <th>max</th> </tbody> <tbody></tbody>

300.000000	300.000000	300.000000	300.000000	300.000000	300.000000	300.000000	300.000000	300.000000	
150.500000	5.326667	2307.386667	1.330000	1468.796667	1986.163333	1.523333	2.926667	8969.453333	2
86.746758	2.873001	1042.561303	1.109898	672.333705	21.377089	1.131543	1.456604	3753.531132	
1.000000	1.000000	504.000000	0.000000	303.000000	1950.000000	0.000000	1.000000	2009.000000	
75.750000	3.000000	1392.250000	0.000000	903.000000	1967.000000	0.000000	2.000000	5996.250000	1
150.500000	5.000000	2265.500000	1.000000	1502.000000	1986.000000	2.000000	3.000000	9031.000000	2
225.250000	8.000000	3306.500000	2.000000	2129.500000	2004.250000	3.000000	4.000000	12316.000000	3
300.000000	10.000000	3998.000000	3.000000	2492.000000	2023.000000	3.000000	5.000000	14987.000000	4

In [4]:
```
df.nunique()
```

Out[4]:
```
Id             300
OverallQual     10
GrLivArea      287
GarageCars       4
TotalBsmtSF    278
YearBuilt       73
FullBath         4
BedroomAbvGr     5
LotArea        296
SalePrice      300
dtype: int64
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
Id             0
OverallQual    0
GrLivArea      0
GarageCars     0
TotalBsmtSF    0
YearBuilt      0
FullBath       0
BedroomAbvGr   0
LotArea        0
SalePrice      0
dtype: int64
```

In [6]:
```
x = df[['OverallQual', 'GrLivArea', 'GarageCars', 'TotalBsmtSF', 'YearBuilt', 'FullBath', 'BedroomAbvGr', 'LotA
y = df[['SalePrice']]
print(x.head())
print(y.head())
```

```
   OverallQual  GrLivArea  GarageCars  TotalBsmtSF  YearBuilt  FullBath  \
0            7       1560           0         1658       1969         2
1            4       2827           2         1319       2012         3
2            8       3920           0          841       2010         1
3            5       3044           0         1058       1998         0
4            7        801           1         2428       2020         0

   BedroomAbvGr  LotArea
0             1     8059
1             4    13530
2             4     9010
3             4    13207
4             1     9117
   SalePrice
0     177106
1     301044
2     360609
3     240556
4     193656
```

In [7]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=12)
```

In [8]:
```
from sklearn.linear_model import LinearRegression
slr=LinearRegression()
slr.fit(x_train,y_train)
slr.predict(x_test)
```

Out[8]:
```
array([[322282.54318338],
       [385458.15265492],
       [207237.33477463],
       [330855.22338302],
       [121908.96253397],
       [375108.10558675],
       [314546.53143915],
       [208021.40165099],
       [195653.48014183],
       [263678.98794911],
       [301812.54037438],
       [193196.30280168],
       [288125.8293911 ],
       [287327.13221277],
       [279901.60413631],
       [259500.11596117],
       [244544.14681399],
       [ 92726.42536953],
       [195930.78396676],
       [380829.69796295],
       [186627.69311552],
       [226935.83169136],
       [236240.71994046],
       [163129.99885132],
       [253733.72020922],
       [251503.62688524],
       [209578.90743169],
       [330829.30525548],
       [174989.22697724],
       [165515.57350818],
       [169754.92556338],
       [219836.21560035],
       [242926.15981885],
       [219144.61969513],
       [395918.92166136],
       [252044.24341116],
       [280225.9308696 ],
       [378973.90634702],
       [148145.86406382],
       [317629.90704933],
       [302248.44565522],
       [143897.85285477],
       [169141.58142887],
       [315231.93304144],
       [216759.68204754],
       [297999.35376511],
       [257229.93545589],
       [244289.262223  ],
       [340693.30193847],
       [368655.09251521],
       [194289.1083612 ],
       [269220.63257098],
       [192068.90251096],
       [210628.44545573],
       [292775.83192996],
       [115431.99476815],
       [206191.5994072 ],
       [282692.11991387],
       [118876.94713104],
       [169476.93187991]])
```

In [9]:
```
from sklearn.metrics import r2_score
y_pred = slr.predict(x_test)
r2 = r2_score(y_test,y_pred)
print(r2)
```

```
0.9776064358983859
```

In [ ]: